

Discrete Probability Distributions

6



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▲ **RECENT STATISTICS SUGGEST** that 15% of those who visit a retail site on the Web make a purchase. A retailer wished to verify this claim. To do so, she selected a sample of 16 “hits” to her site and found that 4 had actually made a purchase. What is the likelihood of exactly four purchases? How many purchases should she expect? What is the likelihood that four or more “hits” result in a purchase? (See Exercise 49 and **LO6-4**.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

- LO6-1** Identify the characteristics of a probability distribution.
- LO6-2** Distinguish between discrete and continuous random variables.
- LO6-3** Compute the mean, variance, and standard deviation of a discrete probability distribution.
- LO6-4** Explain the assumptions of the binomial distribution and apply it to calculate probabilities.
- LO6-5** Explain the assumptions of the hypergeometric distribution and apply it to calculate probabilities.
- LO6-6** Explain the assumptions of the Poisson distribution and apply it to calculate probabilities.

INTRODUCTION

Chapters 2 through 4 are devoted to descriptive statistics. We describe raw data by organizing the data into a frequency distribution and portraying the distribution in tables, graphs, and charts. Also, we compute a measure of location—such as the arithmetic mean, median, or mode—to locate a typical value near the center of the distribution. The range and the standard deviation are used to describe the spread in the data. These chapters focus on describing *something that has already happened*.

Starting with Chapter 5, the emphasis changes—we begin examining *something that could happen*. We note that this facet of statistics is called *statistical inference*. The objective is to make inferences (statements) about a population based on a number of observations, called a sample, selected from the population. In Chapter 5, we state that a probability is a value between 0 and 1 inclusive, and we examine how probabilities can be combined using rules of addition and multiplication.

This chapter begins the study of **probability distributions**. A probability distribution is like a relative frequency distribution. However, instead of describing the past, it is used to provide estimates of the likelihood of future events. Probability distributions can be described by measures of location and dispersion so we show how to compute a distribution's mean, variance, and standard deviation. We also discuss three frequently occurring discrete probability distributions: the binomial, hypergeometric, and Poisson.

LO6-1

Identify the characteristics of a probability distribution.

WHAT IS A PROBABILITY DISTRIBUTION?

A probability distribution defines or describes the likelihoods for a range of possible future outcomes. For example, Spalding Golf Products, Inc. assembles golf clubs with three components: a club head, a shaft, and a grip. From experience five percent of the shafts received from their Asian supplier are defective. As part of Spalding's statistical process control they inspect twenty shafts from each arriving shipment. From experience, we know that the probability of a defective shaft is five percent. Therefore, in a sample of twenty shafts, we would expect one shaft to be defective and the other nineteen shafts to be acceptable. But, by using a probability distribution we can completely describe the range of possible outcomes. For example, we would know the probability that none of the twenty shafts are defective, or that two, or three, or four, or continuing up to twenty shafts in the sample are defective. Given the small probability of a defective shaft, the probability distribution would show that there is a very small probability of four or more defective shafts.

PROBABILITY DISTRIBUTION A listing of all the outcomes of an experiment and the probability associated with each outcome.

The important characteristics of a probability distribution are:

CHARACTERISTICS OF A PROBABILITY DISTRIBUTION

1. The probability of a particular outcome is between 0 and 1 inclusive.
2. The outcomes are mutually exclusive.
3. The list of outcomes is exhaustive. So the sum of the probabilities of the outcomes is equal to 1.

How can we generate a probability distribution? The following example will explain.

▶ **EXAMPLE**

Suppose we are interested in the number of heads showing face up on three tosses of a coin. This is the experiment. The possible results are zero heads, one head, two heads, and three heads. What is the probability distribution for the number of heads?

SOLUTION

There are eight possible outcomes. A tail might appear face up on the first toss, another tail on the second toss, and another tail on the third toss of the coin. Or we might get a tail, tail, and head, in that order. We use the multiplication formula for counting outcomes (5–8). There are $(2)(2)(2)$ or 8 possible results. These results are shown in the following table.

Possible Result	Coin Toss			Number of Heads
	First	Second	Third	
1	T	T	T	0
2	T	T	H	1
3	T	H	T	1
4	T	H	H	2
5	H	T	T	1
6	H	T	H	2
7	H	H	T	2
8	H	H	H	3

Note that the outcome “zero heads” occurred only once, “one head” occurred three times, “two heads” occurred three times, and the outcome “three heads” occurred only once. That is, “zero heads” happened one out of eight times. Thus, the probability of zero heads is one-eighth, the probability of one head is three-eighths, and so on. The probability distribution is shown in Table 6–1. Because one of these outcomes must happen, the total of the probabilities of all possible events is 1.000. This is always true. The same information is shown in Chart 6–1.

TABLE 6–1 Probability Distribution for the Events of Zero, One, Two, and Three Heads Showing Face Up on Three Tosses of a Coin

Number of Heads, x	Probability of Outcome, $P(x)$
0	$\frac{1}{8} = .125$
1	$\frac{3}{8} = .375$
2	$\frac{3}{8} = .375$
3	$\frac{1}{8} = .125$
Total	$\frac{8}{8} = 1.000$

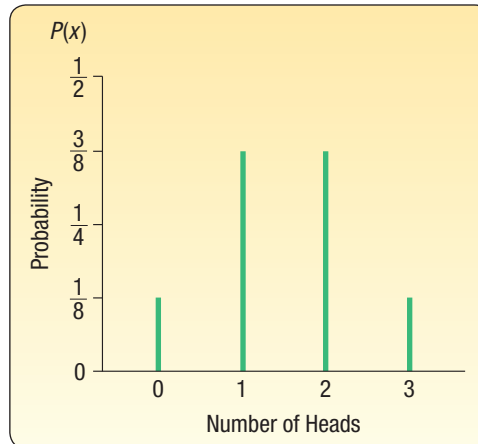


CHART 6-1 Graphical Presentation of the Number of Heads Resulting from Three Tosses of a Coin and the Corresponding Probability

Refer to the coin-tossing example in Table 6-1. We write the probability of x as $P(x)$. So the probability of zero heads is $P(0 \text{ heads}) = .125$, and the probability of one head is $P(1 \text{ head}) = .375$, and so forth. The sum of these mutually exclusive probabilities is 1; that is, from Table 6-1, $.125 + .375 + .375 + .125 = 1.00$.

SELF-REVIEW 6-1



The possible outcomes of an experiment involving the roll of a six-sided die are a one-spot, a two-spot, a three-spot, a four-spot, a five-spot, and a six-spot.

- Develop a probability distribution for the number of possible spots.
- Portray the probability distribution graphically.
- What is the sum of the probabilities?

LO6-2

Distinguish between discrete and continuous random variables.

RANDOM VARIABLES

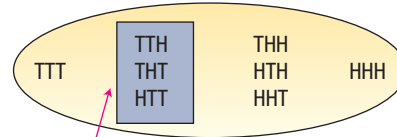
In any experiment of chance, the outcomes occur randomly. So it is often called a *random variable*. For example, rolling a single die is an experiment: Any one of six possible outcomes can occur. Some experiments result in outcomes that are measured with quantitative variables (such as dollars, weight, or number of children), and other experimental outcomes are measured with qualitative variables (such as color or religious preference). A few examples will further illustrate what is meant by a **random variable**.

- The number of employees absent from the day shift on Monday, the number might be 0, 1, 2, 3, . . . The number absent is the random variable.
- The hourly wage of a sample of 50 plumbers in Jacksonville, FL. The hourly wage is the random variable.
- The number of defective lightbulbs produced in an hour at the Cleveland Electric Company, Inc.
- The grade level (Freshman, Sophomore, Junior, or Senior) of the members of the St. James High School Varsity girls' basketball team. The grade level is the random variable and notice that it is a qualitative variable.
- The number of participants in the 2016 New York City Marathon.
- The daily number of drivers charged with driving under the influence of alcohol in Brazoria County, Texas, last month.

A random variable is defined as follows:

RANDOM VARIABLE A variable measured or observed as the result of an experiment. By chance, the variable can have different values.

In Chapter 5 we defined the terms *experiment*, *outcome*, and *event*. Consider the example we just described regarding the experiment of tossing a fair coin three times. In this case the *random variable* is the number of heads that appear in the three tosses. There are eight possible outcomes to this experiment. These outcomes are shown in the following diagram.

Possible *outcomes* for three coin tosses

The *event* {one head} occurs and the *random variable* $x = 1$.

So, one possible outcome is that a tail appears on each toss: TTT. This single outcome would describe the event of zero heads appearing in three tosses. Another possible outcome is a head followed by two tails: HTT. If we wish to determine the event of exactly one head appearing in the three tosses, we must consider the three possible outcomes: TTH, THT, and HTT. These three outcomes describe the event of exactly one head appearing in three tosses.

In this experiment, the random variable is the number of heads in three tosses. The random variable can have four different values, 0, 1, 2, or 3. The outcomes of the experiment are unknown. But, using probability, we can compute the probability of a single head in three tosses as $3/8$ or 0.375. As shown in Chapter 5, the probability of each value of the random variable can be computed to create a probability distribution for the random variable, number of heads in three tosses of a coin.

There are two types of random variables: *discrete* or *continuous*.

Discrete Random Variable

A discrete random variable can assume only a certain number of separated values. For example, the Bank of the Carolinas counts the number of credit cards carried for a group of customers. The data are summarized with the following relative frequency table.

Number of Credit Cards	Relative Frequency
0	.03
1	.10
2	.18
3	.21
4 or more	.48
Total	1.00

In this frequency table, the number of cards carried is the **discrete random variable**.

DISCRETE RANDOM VARIABLE A random variable that can assume only certain clearly separated values.

A discrete random variable can, in some cases, assume fractional or decimal values. To be a discrete random variable, these values must be separated—that is, have distance between them. As an example, a department store offers coupons with discounts of 10%, 15%, and 25%. In terms of probability, we could compute the probability that a customer would use a 10% coupon versus a 15% or 25% coupon.

Continuous Random Variable

On the other hand, a **continuous random variable** can assume an infinite number of values within a given range. It is measured on a continuous interval or ratio scale. Examples include:

- The times of commercial flights between Atlanta and Los Angeles are 4.67 hours, 5.13 hours, and so on. The random variable is the time in hours and is measured on a continuous scale of time.
- The annual snowfall in Minneapolis, Minnesota. The random variable is the amount of snow, measured on a continuous scale.

As with discrete random variables, the likelihood of a continuous random variable can be summarized with a **probability distribution**. For example, with a probability distribution for the flight time between Atlanta and Los Angeles, we could say that there is a probability of 0.90 that the flight will be less than 4.5 hours. This also implies that there is a probability of 0.10 that the flight will be more than 4.5 hours. With a probability of snowfall in Minneapolis, we could say that there is probability of 0.25 that the annual snowfall will exceed 48 inches. This also implies that there is a probability of 0.75 that annual snowfall will be less than 48 inches. Notice that these examples refer to a continuous range of values.

LO6-3

Compute the mean, variance, and standard deviation of a probability distribution.

THE MEAN, VARIANCE, AND STANDARD DEVIATION OF A DISCRETE PROBABILITY DISTRIBUTION

In Chapter 3, we discussed measures of location and variation for a frequency distribution. The mean reports the central location of the data, and the variance describes the spread in the data. In a similar fashion, a probability distribution is summarized by its mean and variance. We identify the mean of a probability distribution by the lowercase Greek letter mu (μ) and the standard deviation by the lowercase Greek letter sigma (σ).

Mean

The mean is a typical value used to represent the central location of a probability distribution. It also is the long-run average value of the random variable. The mean of a probability distribution is also referred to as its **expected value**. It is a weighted average where the possible values of a random variable are weighted by their corresponding probabilities of occurrence.

The mean of a discrete probability distribution is computed by the formula:

MEAN OF A PROBABILITY DISTRIBUTION

$$\mu = \sum[xP(x)]$$

(6-1)

where $P(x)$ is the probability of a particular value x . In other words, multiply each x value by its probability of occurrence, and then add these products.

Variance and Standard Deviation

The mean is a typical value used to summarize a discrete probability distribution. However, it does not describe the amount of spread (variation) in a distribution. The variance does this. The formula for the variance of a probability distribution is:

VARIANCE OF A PROBABILITY DISTRIBUTION

$$\sigma^2 = \sum[(x - \mu)^2 P(x)]$$

(6-2)

The computational steps are:

1. Subtract the mean from each value of the random variable, and square this difference.
2. Multiply each squared difference by its probability.
3. Sum the resulting products to arrive at the variance.

The standard deviation, σ , is found by taking the positive square root of σ^2 ; that is,
 $\sigma = \sqrt{\sigma^2}$

An example will help explain the details of the calculation and interpretation of the mean and standard deviation of a probability distribution.

EXAMPLE



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John Ragsdale sells new cars for Pelican Ford. John usually sells the largest number of cars on Saturday. He has developed the following probability distribution for the number of cars he expects to sell on a particular Saturday.

Number of Cars Sold, x	Probability, $P(x)$
0	.1
1	.2
2	.3
3	.3
4	.1
	<u>1.0</u>

1. What type of distribution is this?
2. On a typical Saturday, how many cars does John expect to sell?
3. What is the variance of the distribution?

SOLUTION

1. This is a discrete probability distribution for the random variable called “number of cars sold.” Note that John expects to sell only within a certain range of cars; he does not expect to sell 5 cars or 50 cars. Further, he cannot sell half a car. He can sell only 0, 1, 2, 3, or 4 cars. Also, the outcomes are mutually exclusive—he cannot sell a total of both 3 and 4 cars on the same Saturday. The sum of the possible outcomes total 1. Hence, these circumstance qualify as a probability distribution.
2. The mean number of cars sold is computed by weighting the number of cars sold by the probability of selling that number and adding or summing the products, using formula (6–1):

$$\begin{aligned}
 \mu &= \Sigma[xP(x)] \\
 &= 0(.1) + 1(.2) + 2(.3) + 3(.3) + 4(.1) \\
 &= 2.1
 \end{aligned}$$

These calculations are summarized in the following table.

Number of Cars Sold, x	Probability, $P(x)$	$x \cdot P(x)$
0	.1	0.0
1	.2	0.2
2	.3	0.6
3	.3	0.9
4	.1	0.4
	<u>1.0</u>	<u>$\mu = 2.1$</u>

How do we interpret a mean of 2.1? This value indicates that, over a large number of Saturdays, John Ragsdale expects to sell a mean of 2.1 cars a day. Of

course, it is not possible for him to sell *exactly* 2.1 cars on any particular Saturday. However, the expected value can be used to predict the arithmetic mean number of cars sold on Saturdays in the long run. For example, if John works 50 Saturdays during a year, he can expect to sell $(50)(2.1)$ or 105 cars just on Saturdays. Thus, the mean is sometimes called the expected value.

- The following table illustrates the steps to calculate the variance using formula (6-2). The first two columns repeat the probability distribution. In column three, the mean is subtracted from each value of the random variable. In column four, the differences from column three are squared. In the fifth column, each squared difference in column four is multiplied by the corresponding probability. The variance is the sum of the values in column five.

Number of Cars Sold, x	Probability, $P(x)$	$(x - \mu)$	$(x - \mu)^2$	$(x - \mu)^2 P(x)$
0	.1	$0 - 2.1$	4.41	0.441
1	.2	$1 - 2.1$	1.21	0.242
2	.3	$2 - 2.1$	0.01	0.003
3	.3	$3 - 2.1$	0.81	0.243
4	.1	$4 - 2.1$	3.61	0.361
				$\sigma^2 = 1.290$

Recall that the standard deviation, σ , is the positive square root of the variance. In this example, $\sqrt{\sigma^2} = \sqrt{1.290} = 1.136$ cars. How do we apply a standard deviation of 1.136 cars? If salesperson Rita Kirsch also sold a mean of 2.1 cars on Saturdays, and the standard deviation in her sales was 1.91 cars, we would conclude that there is more variability in the Saturday sales of Ms. Kirsch than in those of Mr. Ragsdale (because $1.91 > 1.136$).

SELF-REVIEW 6-2



The Pizza Palace offers three sizes of cola. The smallest size sells for \$1.99, the medium for \$2.49, and the large for \$2.89. Thirty percent of the drinks sold are small, 50% are medium, and 20% are large. Create a probability distribution for the random variable price and answer the following questions.

- Is this a discrete probability distribution? Indicate why or why not.
- Compute the mean amount charged for a cola.
- What is the variance in the amount charged for a cola? The standard deviation?

EXERCISES

- FILE** Compute the mean and variance of the following discrete probability distribution.

x	$P(x)$
0	.2
1	.4
2	.3
3	.1

- FILE** Compute the mean and variance of the following discrete probability distribution.

x	$P(x)$
2	.5
8	.3
10	.2

3. **FILE** Compute the mean and variance of the following probability distribution.

x	$P(x)$
5	.1
10	.3
15	.2
20	.4

4. Which of these variables are discrete and which are continuous random variables?
- The number of new accounts established by a salesperson in a year.
 - The time between customer arrivals to a bank ATM.
 - The number of customers in Big Nick's barber shop.
 - The amount of fuel in your car's gas tank.
 - The number of minorities on a jury.
 - The outside temperature today.
5. **FILE** The information below is the number of daily emergency service calls made by the volunteer ambulance service of Walterboro, South Carolina, for the last 50 days. To explain, there were 22 days on which there were two emergency calls, and 9 days on which there were three emergency calls.

Number of Calls	Frequency
0	8
1	10
2	22
3	9
4	1
Total	50

- Convert this information on the number of calls to a probability distribution.
 - Is this an example of a discrete or continuous probability distribution?
 - What is the mean number of emergency calls per day?
 - What is the standard deviation of the number of calls made daily?
6. **FILE** The director of admissions at Kinzua University in Nova Scotia estimated the distribution of student admissions for the fall semester on the basis of past experience. What is the expected number of admissions for the fall semester? Compute the variance and the standard deviation of the number of admissions.

Admissions	Probability
1,000	.6
1,200	.3
1,500	.1

7. **FILE** Belk Department Store is having a special sale this weekend. Customers charging purchases of more than \$50 to their Belk credit card will be given a special Belk Lottery card. The customer will scratch off the card, which will indicate the amount to be taken off the total amount of the purchase. Listed below are the amount of the prize and the percent of the time that amount will be deducted from the total amount of the purchase.

Prize Amount	Probability
\$ 10	.50
25	.40
50	.08
100	.02

- a. What is the mean amount deducted from the total purchase amount?
- b. What is the standard deviation of the amount deducted from the total purchase?
8. **FILE** The Downtown Parking Authority of Tampa, Florida, reported the following information for a sample of 250 customers on the number of hours cars are parked and the amount they are charged.

Number of Hours	Frequency	Amount Charged
1	20	\$ 3
2	38	6
3	53	9
4	45	12
5	40	14
6	13	16
7	5	18
8	36	20
	250	

- a. Convert the information on the number of hours parked to a probability distribution. Is this a discrete or a continuous probability distribution?
- b. Find the mean and the standard deviation of the number of hours parked. How would you answer the question: How long is a typical customer parked?
- c. Find the mean and the standard deviation of the amount charged.

LO6-4

Explain the assumptions of the binomial distribution and apply it to calculate probabilities.

BINOMIAL PROBABILITY DISTRIBUTION

The **binomial probability distribution** is a widely occurring discrete probability distribution. To describe experimental outcomes with a binomial distribution, there are four requirements. The first requirement is there are only two possible outcomes on a particular experimental trial. For example, on a test, a true/false question is either answered correctly or incorrectly. In a resort, a housekeeping supervisor reviews an employee's work and evaluates it as acceptable or unacceptable. A key characteristic of the two outcomes is that they must be mutually exclusive. This means that the answer to a true/false question must be either correct or incorrect but cannot be both correct and incorrect at the same time. Another example is the outcome of a sales call. Either a customer purchases or does not purchase the product, but the sale cannot result in both outcomes. Frequently, we refer to the two possible outcomes of a binomial experiment as a "success" and a "failure." However, this distinction does not imply that one outcome is good and the other is bad, only that there are two mutually exclusive outcomes.

The second binomial requirement is that the random variable is the number of successes for a fixed and known number of trials. For example, we flip a coin five times and count the number of times a head appears in the five flips, we randomly select 10 employees and count the number who are older than 50 years of age, or we randomly select 20 boxes of Kellogg's Raisin Bran and count the number that weigh more than the amount indicated on the package. In each example, we count the number of successes from the fixed number of trials.

A third requirement is that we know the probability of a success and it is the same for each trial. Three examples are:

- For a test with 10 true/false questions, we know there are 10 trials and the probability of correctly guessing the answer for any of the 10 trials is 0.5. Or, for a test with 20 multiple-choice questions with four options and only one correct answer, we know that there are 20 trials and the probability of randomly guessing the correct answer for each of the 20 trials is 0.25.

- Bones Albaugh is a Division I college basketball player who makes 70% of his foul shots. If he has five opportunities in tonight's game, the likelihood he will be successful on each of the five attempts is 0.70.
- In a recent poll, 18% of adults indicated a Snickers bar was their favorite candy bar. We select a sample of 15 adults and ask each for his or her favorite candy bar. The likelihood a Snickers bar is the answer for each adult is 0.18.



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The final requirement of a binomial probability distribution is that each trial is *independent* of any other trial. Independent means there is no pattern to the trials. The outcome of a particular trial does not affect the outcome of any other trial. Two examples are:

- A young family has two children, both boys. The probability of a third birth being a boy is still .50. That is, the gender of the third child is independent of the gender of the other two.
- Suppose 20% of the patients served in the emergency room at Waccamaw Hospital do not have insurance. If the second patient served on the afternoon shift today did not have insurance, that does not affect the probability the third, the tenth, or any of the other patients will or will not have insurance.

BINOMIAL PROBABILITY EXPERIMENT

1. An outcome on each trial of an experiment is classified into one of two mutually exclusive categories—a success or a failure.
2. The random variable is the number of successes in a fixed number of trials.
3. The probability of success is the same for each trial.
4. The trials are independent, meaning that the outcome of one trial does not affect the outcome of any other trial.

How Is a Binomial Probability Computed?

To construct a particular binomial probability, we use (1) the number of trials and (2) the probability of success on each trial. For example, if the Hannah Landscaping Company plants 10 Norfolk pine trees today knowing that 90% of these trees survive, we can compute the binomial probability that exactly 8 trees survive. In this case the number of trials is the 10 trees, the probability of success is .90, and the number of successes is eight. In fact, we can compute a binomial probability for any number of successes from 0 to 10 surviving trees.

A binomial probability is computed by the formula:

BINOMIAL PROBABILITY FORMULA

$$P(x) = {}_nC_x \pi^x (1 - \pi)^{n-x} \quad (6-3)$$

where:

C denotes a combination.

n is the number of trials.

x is the random variable defined as the number of successes.

π is the probability of a success on each trial.

We use the Greek letter π (pi) to denote a binomial population parameter. Do not confuse it with the mathematical constant 3.1416.

EXAMPLE

There are five flights daily from Pittsburgh via American Airlines into the Bradford Regional Airport in Bradford, Pennsylvania. Suppose the probability that any flight arrives late is .20. What is the probability that none of the flights are late today? What is the probability that exactly one of the flights is late today?

SOLUTION

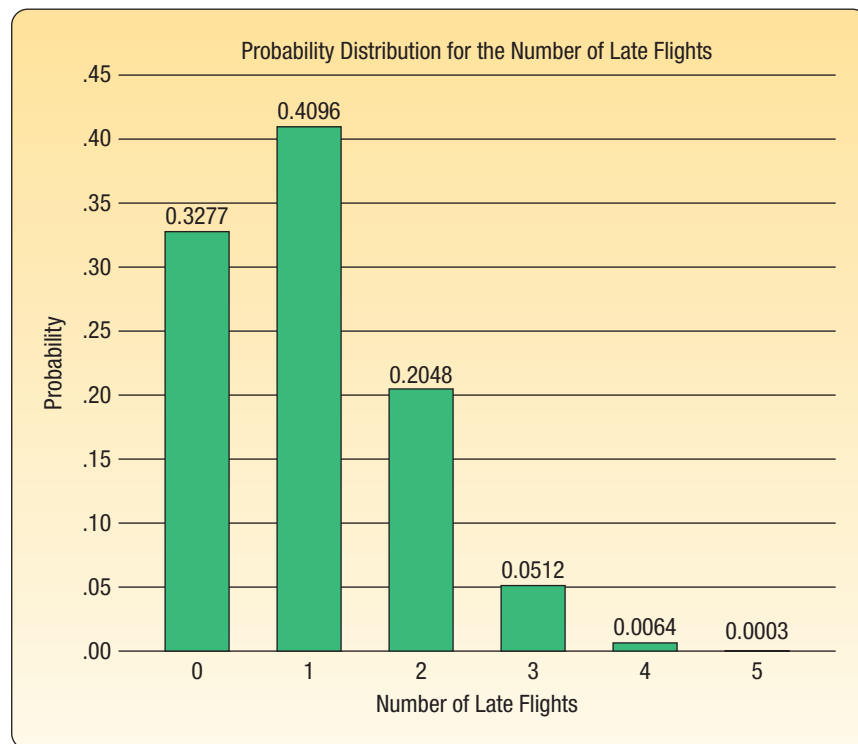
We can use formula (6–3). The probability that a particular flight is late is .20, so let $\pi = .20$. There are five flights, so $n = 5$, and X , the random variable, refers to the number of successes. In this case, a “success” is a flight that arrives late. The random variable, x , can be equal to 0 late flights in the five trials, 1 late flight in the five trials, or 2, 3, 4, or 5. The probability for no late arrivals, $x = 0$, is,

$$\begin{aligned} P(0) &= {}_n C_x (\pi)^x (1 - \pi)^{n-x} \\ &= {}_5 C_0 (.20)^0 (1 - .20)^{5-0} = (1)(1)(.3277) = .3277 \end{aligned}$$

The probability that exactly one of the five flights will arrive late today is .4096, found by

$$\begin{aligned} P(1) &= {}_n C_x (\pi)^x (1 - \pi)^{n-x} \\ &= {}_5 C_1 (.20)^1 (1 - .20)^{5-1} = (5)(.20)(.4096) = .4096 \end{aligned}$$

The entire binomial probability distribution with $\pi = .20$ and $n = 5$ is shown in the following bar chart. We observe that the probability of exactly three late flights is .0512 and from the bar chart that the distribution of the number of late arrivals is positively skewed.



The mean (μ) and the variance (σ^2) of a binomial distribution are computed in a “shortcut” fashion by:

MEAN OF A BINOMIAL DISTRIBUTION

$$\mu = n\pi$$

(6–4)**VARIANCE OF A BINOMIAL DISTRIBUTION**

$$\sigma^2 = n\pi(1 - \pi)$$

(6–5)

For the example regarding the number of late flights, recall that $\pi = .20$ and $n = 5$. Hence:

$$\mu = n\pi = (5)(.20) = 1.0$$

$$\sigma^2 = n\pi(1 - \pi) = 5(.20)(1 - .20) = .7997$$

The mean of 1.0 and the variance of .7997 can be verified from formulas (6–1) and (6–2). The probability distribution from the bar chart shown earlier and the details of the calculations are shown below.

Number of Late Flights,					
x	$P(x)$	$xP(x)$	$x - \mu$	$(x - \mu)^2$	$(x - \mu)^2P(x)$
0	0.3277	0.0000	–1	1	0.3277
1	0.4096	0.4096	0	0	0
2	0.2048	0.4096	1	1	0.2048
3	0.0512	0.1536	2	4	0.2048
4	0.0064	0.0256	3	9	0.0576
5	0.0003	0.0015	4	16	0.0048
		$\mu = 1.0000$	$\sigma^2 = 0.7997$		

Binomial Probability Tables

Formula (6–3) can be used to build a binomial probability distribution for any value of n and π . However, for a larger n , the calculations take more time. For convenience, the tables in Appendix B.1 show the result of using the formula for various values of n and π . Table 6–2 shows part of Appendix B.1 for $n = 6$ and various values of π .

TABLE 6–2 Binomial Probabilities for $n = 6$ and Selected Values of π

$n = 6$ Probability											
$x \backslash \pi$	.05	.1	.2	.3	.4	.5	.6	.7	.8	.9	.95
0	.735	.531	.262	.118	.047	.016	.004	.001	.000	.000	.000
1	.232	.354	.393	.303	.187	.094	.037	.010	.002	.000	.000
2	.031	.098	.246	.324	.311	.234	.138	.060	.015	.001	.000
3	.002	.015	.082	.185	.276	.313	.276	.185	.082	.015	.002
4	.000	.001	.015	.060	.138	.234	.311	.324	.246	.098	.031
5	.000	.000	.002	.010	.037	.094	.187	.303	.393	.534	.232
6	.000	.000	.000	.001	.004	.016	.047	.118	.262	.531	.735

EXAMPLE

In the Southwest, 5% of all cell phone calls are dropped. What is the probability that out of six randomly selected calls, none was dropped? Exactly one? Exactly two? Exactly three? Exactly four? Exactly five? Exactly six out of six?

SOLUTION

The binomial conditions are met: (a) there are only two possible outcomes (a particular call is either dropped or not dropped), (b) there are a fixed number of trials (6), (c) there is a constant probability of success (.05), and (d) the trials are independent.

Refer to Table 6–2 on the previous page for the probability of exactly zero dropped calls. Go down the left margin to an x of 0. Now move horizontally to the column headed by a π of .05 to find the probability. It is .735. The values in Table 6–2 are rounded to three decimal places.

The probability of exactly one dropped call in a sample of six calls is .232. The complete binomial probability distribution for $n = 6$ and $\pi = .05$ is:

Number of Dropped Calls, x	Probability of Occurrence, $P(x)$	Number of Dropped Calls, x	Probability of Occurrence, $P(x)$
0	.735	4	.000
1	.232	5	.000
2	.031	6	.000
3	.002		

Of course, there is a slight chance of getting exactly five dropped calls out of six random selections. It is .00000178, found by inserting the appropriate values in the binomial formula:

$$P(5) = {}_6C_5(.50)^5(.95)^1 = (6)(.05)^5(.95) = .00000178$$

For six out of the six, the exact probability is .000000016. Thus, the probability is very small that five or six calls will be dropped in six trials.

We can compute the mean or expected value of the distribution of the number defective:

$$\begin{aligned}\mu &= n\pi = (6)(.05) = 0.30 \\ \sigma^2 &= n\pi(1 - \pi) = 6(.05)(.95) = 0.285\end{aligned}$$

SELF-REVIEW 6-3



Ninety-five percent of the employees at the J. M. Smucker Company plant on Laskey Road have their bimonthly wages sent directly to their bank by electronic funds transfer. This is also called direct deposit. Suppose we select a random sample of seven employees.

- (a) Does this situation fit the assumptions of the binomial distribution?
- (b) What is the probability that all seven employees use direct deposit?
- (c) Use formula (6–3) to determine the exact probability that four of the seven sampled employees use direct deposit.
- (d) Use Excel to verify your answers to parts (b) and (c).

Appendix B.1 is limited. It gives probabilities for n values from 1 to 15 and π values of .05, .10, . . . , .90, and .95. A software program can generate the probabilities for a specified number of successes, given n and π . The Excel output on the next page shows the probability when $n = 40$ and $\pi = .09$. Note that the number of successes stops at 15 because the probabilities for 16 to 40 are very close to 0. The instructions are detailed in the Software Commands in Appendix C.

	A	B
1	Success	Probability
2	0	0.0230
3	1	0.0910
4	2	0.1754
5	3	0.2198
6	4	0.2011
7	5	0.1432
8	6	0.0826
9	7	0.0397
10	8	0.0162
11	9	0.0057
12	10	0.0017
13	11	0.0005
14	12	0.0001
15	13	0.0000
16	14	0.0000
17	15	0.0000

Several additional points should be made regarding the binomial probability distribution.

1. If n remains the same but π increases from .05 to .95, the shape of the distribution changes. Look at Table 6–3 and Chart 6–2. The distribution for a π of .05 is positively skewed. As π approaches .50, the distribution becomes symmetrical. As π goes beyond .50 and moves toward .95, the probability distribution becomes negatively skewed. Table 6–3 highlights probabilities for $n = 10$ and a π of .05, .10, .20, .50, and .70. The graphs of these probability distributions are shown in Chart 6–2.

TABLE 6–3 Probability of 0, 1, 2, . . . Successes for a π of .05, .10, .20, .50, and .70, and an n of 10

$x \backslash \pi$	.05	.1	.2	.3	.4	.5	.6	.7	.8	.9	.95
0	.599	.349	.107	.028	.006	.001	.000	.000	.000	.000	.000
1	.315	.387	.268	.121	.040	.010	.002	.000	.000	.000	.000
2	.075	.194	.302	.233	.121	.044	.011	.001	.000	.000	.000
3	.010	.057	.201	.267	.215	.117	.042	.009	.001	.000	.000
4	.001	.011	.088	.200	.251	.205	.111	.037	.006	.000	.000
5	.000	.001	.026	.103	.201	.246	.201	.103	.026	.001	.000
6	.000	.000	.006	.037	.111	.205	.251	.200	.088	.011	.001
7	.000	.000	.001	.009	.042	.117	.215	.267	.201	.057	.010
8	.000	.000	.000	.001	.011	.044	.121	.233	.302	.194	.075
9	.000	.000	.000	.000	.002	.010	.040	.121	.268	.387	.315
10	.000	.000	.000	.000	.000	.001	.006	.028	.107	.349	.599

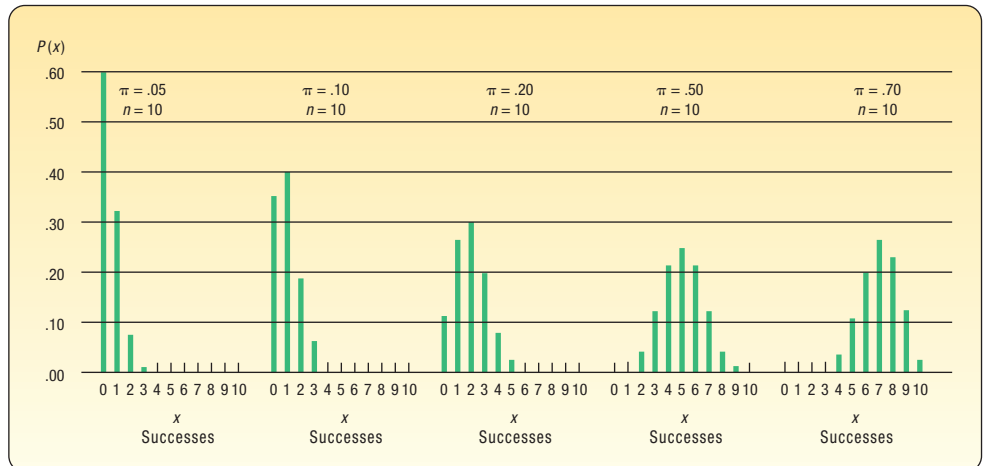


CHART 6–2 Graphing the Binomial Probability Distribution for a π of .05, .10, .20, .50, and .70, and an n of 10

2. If π , the probability of success, remains the same but n becomes larger, the shape of the binomial distribution becomes more symmetrical. Chart 6–3 shows a situation where π remains constant at .10 but n increases from 7 to 40.

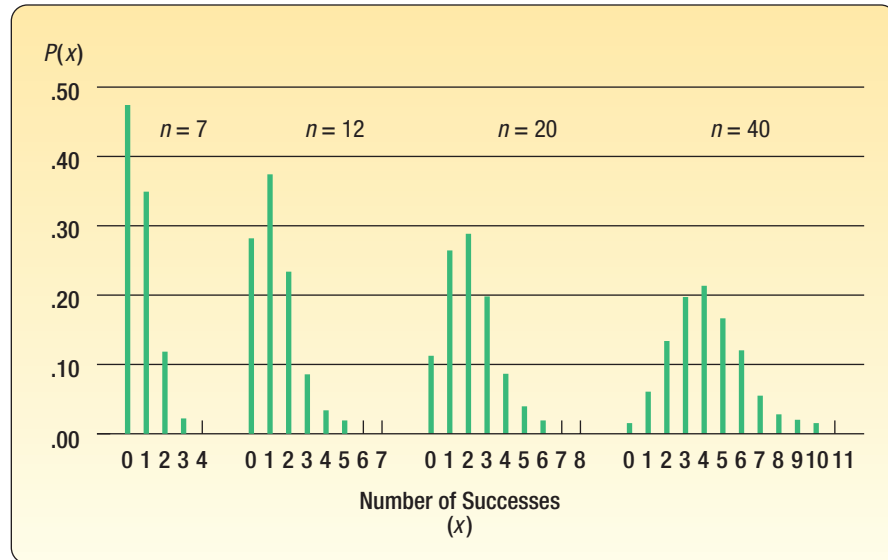


CHART 6-3 Chart Representing the Binomial Probability Distribution for a π of .10 and an n of 7, 12, 20, and 40

EXERCISES

9. In a binomial situation, $n = 4$ and $\pi = .25$. Determine the probabilities of the following events using the binomial formula.
 - a. $x = 2$
 - b. $x = 3$
10. In a binomial situation, $n = 5$ and $\pi = .40$. Determine the probabilities of the following events using the binomial formula.
 - a. $x = 1$
 - b. $x = 2$
11. Assume a binomial distribution where $n = 3$ and $\pi = .60$.
 - a. Refer to Appendix B.1, and list the probabilities for values of x from 0 to 3.
 - b. Determine the mean and standard deviation of the distribution from the general definitions given in formulas (6-1) and (6-2).
12. Assume a binomial distribution where $n = 5$ and $\pi = .30$.
 - a. Refer to Appendix B.1 and list the probabilities for values of x from 0 to 5.
 - b. Determine the mean and standard deviation of the distribution from the general definitions given in formulas (6-1) and (6-2).
13. An American Society of Investors survey found 30% of individual investors have used a discount broker. In a random sample of nine individuals, what is the probability:
 - a. Exactly two of the sampled individuals have used a discount broker?
 - b. Exactly four of them have used a discount broker?
 - c. None of them has used a discount broker?
14. **FILE** The U.S. Postal Service reports 95% of first-class mail within the same city is delivered within 2 days of the time of mailing. Six letters are randomly sent to different locations.
 - a. What is the probability that all six arrive within 2 days?
 - b. What is the probability that exactly five arrive within 2 days?
 - c. Find the mean number of letters that will arrive within 2 days.
 - d. Compute the variance and standard deviation of the number that will arrive within 2 days.

15. **FILE** Industry standards suggest that 10% of new vehicles require warranty service within the first year. Jones Nissan in Sumter, South Carolina, sold 12 Nissans yesterday.
 - a. What is the probability that none of these vehicles requires warranty service?
 - b. What is the probability exactly one of these vehicles requires warranty service?
 - c. Determine the probability that exactly two of these vehicles require warranty service.
 - d. Compute the mean and standard deviation of this probability distribution.
16. **FILE** A telemarketer makes six phone calls per hour and is able to make a sale on 30% of these contacts. During the next 2 hours, find:
 - a. The probability of making exactly four sales.
 - b. The probability of making no sales.
 - c. The probability of making exactly two sales.
 - d. The mean number of sales in the 2-hour period.
17. **FILE** A recent survey by the American Accounting Association revealed 23% of students graduating with a major in accounting select public accounting. Suppose we select a sample of 15 recent graduates.
 - a. What is the probability two select public accounting?
 - b. What is the probability five select public accounting?
 - c. How many graduates would you expect to select public accounting?
18. **FILE** It is reported that 41% of American households use a cell phone exclusively for their telephone service. In a sample of eight households,
 - a. Find the probability that no household uses a cell phone as their exclusive telephone service.
 - b. Find the probability that exactly 5 households exclusively use a cell phone for telephone service.
 - c. Find the mean number of households exclusively using cell phones.

Cumulative Binomial Probability Distributions

We may wish to know the probability of correctly guessing the answers to 6 or more true/false questions out of 10. Or we may be interested in the probability of *selecting less than two* defectives at random from production during the previous hour. In these cases, we need cumulative frequency distributions similar to the ones developed in the Chapter 2, Cumulative Distribution section on page 38. The following example will illustrate.

EXAMPLE

A study by the Illinois Department of Transportation concluded that 76.2% of front seat occupants used seat belts. That is, both occupants of the front seat were using their seat belts. Suppose we decide to compare that information with current usage. We select a sample of 12 vehicles.

1. What is the probability the front seat occupants in exactly 7 of the 12 vehicles selected are wearing seat belts?
2. What is the probability the front seat occupants in at least 7 of the 12 vehicles are wearing seat belts?

SOLUTION

This situation meets the binomial requirements.

- In a particular vehicle, both the front seat occupants are either wearing seat belts or they are not. There are only two possible outcomes.
- There are a fixed number of trials, 12 in this case, because 12 vehicles are checked.
- The probability of a “success” (occupants wearing seat belts) is the same from one vehicle to the next: 76.2%.
- The trials are independent. If the fourth vehicle selected in the sample has all the occupants wearing their seat belts, this does not have any effect on the results for the fifth or tenth vehicle.

To find the likelihood the occupants of *exactly* 7 of the sampled vehicles are wearing seat belts, we use formula (6–3). In this case, $n = 12$ and $\pi = .762$.

$$P(x = 7) = {}_{12}C_7(.762)^7(1 - .762)^{12-7} = 792(.149171)(.000764) = .0902$$

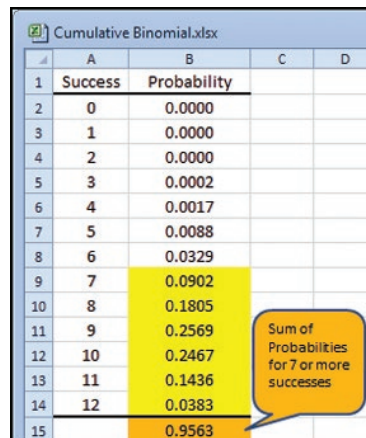
So we conclude the likelihood that the occupants of exactly 7 of the 12 sampled vehicles will be wearing their seat belts is about 9%.

To find the probability that the occupants in seven or more of the vehicles will be wearing seat belts, we use formula (6–3) from this chapter as well as the special rule of addition from the previous chapter. See formula (5-2) on page 141.

Because the events are mutually exclusive (meaning that a particular sample of 12 vehicles cannot have both a *total* of 7 and a *total* of 8 vehicles where the occupants are wearing seat belts), we find the probability of 7 vehicles where the occupants are wearing seat belts, the probability of 8, and so on up to the probability that occupants of all 12 sample vehicles are wearing seat belts. The probability of each of these outcomes is then totaled.

$$\begin{aligned} P(x \geq 7) &= P(x = 7) + P(x = 8) + P(x = 9) + P(x = 10) + P(x = 11) + P(x = 12) \\ &= .0902 + .1805 + .2569 + .2467 + .1436 + .0383 \\ &= .9562 \end{aligned}$$

So the probability of selecting 12 cars and finding that the occupants of 7 or more vehicles were wearing seat belts is .9562. This information is shown on the following Excel spreadsheet. There is a slight difference in the software answer due to rounding. The Excel commands are similar to those detailed in the Software Commands in Appendix C.



	A	B	C	D
1	Success	Probability		
2	0	0.0000		
3	1	0.0000		
4	2	0.0000		
5	3	0.0002		
6	4	0.0017		
7	5	0.0088		
8	6	0.0329		
9	7	0.0902		
10	8	0.1805		
11	9	0.2569		
12	10	0.2467		
13	11	0.1436		
14	12	0.0383		
15		0.9563		

SELF-REVIEW 6-4



A recent study revealed that 40% of women in the San Diego metropolitan area who work full time also volunteer in the community. Suppose we randomly select eight women in the San Diego area.

- What are the values for n and π ?
- What is the probability exactly three of the women volunteer in the community?
- What is the probability at least one of the women volunteers in the community?

EXERCISES

- In a binomial distribution, $n = 8$ and $\pi = .30$. Find the probabilities of the following events.
 - $x = 2$.
 - $x \leq 2$ (the probability that x is equal to or less than 2).
 - $x \geq 3$ (the probability that x is equal to or greater than 3).
- In a binomial distribution, $n = 12$ and $\pi = .60$. Find the following probabilities.
 - $x = 5$.
 - $x \leq 5$.
 - $x \geq 6$.
- FILE** In a recent study, 90% of the homes in the United States were found to have large-screen TVs. In a sample of nine homes, what is the probability that:
 - All nine have large-screen TVs?
 - Less than five have large-screen TVs?
 - More than five have large-screen TVs?
 - At least seven homes have large-screen TVs?
- FILE** A manufacturer of window frames knows from long experience that 5% of the production will have some type of minor defect that will require an adjustment. What is the probability that in a sample of 20 window frames:
 - None will need adjustment?
 - At least one will need adjustment?
 - More than two will need adjustment?
- FILE** The speed with which utility companies can resolve problems is very important. GTC, the Georgetown Telephone Company, reports it can resolve customer problems the same day they are reported in 70% of the cases. Suppose the 15 cases reported today are representative of all complaints.
 - How many of the problems would you expect to be resolved today? What is the standard deviation?
 - What is the probability 10 of the problems can be resolved today?
 - What is the probability 10 or 11 of the problems can be resolved today?
 - What is the probability more than 10 of the problems can be resolved today?
- FILE** It is asserted that 80% of the cars approaching an individual toll booth in New Jersey are equipped with an E-ZPass transponder. Find the probability that in a sample of six cars:
 - All six will have the transponder.
 - At least three will have the transponder.
 - None will have a transponder.

LO6-5

Explain the assumptions of the hypergeometric distribution and apply it to calculate probabilities.

HYPERGEOMETRIC PROBABILITY DISTRIBUTION

For the binomial distribution to be applied, the probability of a success must stay the same for each trial. For example, the probability of guessing the correct answer to a true/false question is .50. This probability remains the same for each question on an examination. Likewise, suppose that 40% of the registered voters in a precinct are

Republicans. If 27 registered voters are selected at random, the probability of choosing a Republican on the first selection is .40. The chance of choosing a Republican on the next selection is also .40, assuming that the sampling is done *with replacement*, meaning that the person selected is put back in the population before the next person is selected.

Most sampling, however, is done *without replacement*. Thus, if the population is small, the probability of a success will change for each observation. For example, if the population consists of 20 items, the probability of selecting a particular item from that population is 1/20. If the sampling is done without replacement, after the first selection there are only 19 items remaining; the probability of selecting a particular item on the second selection is only 1/19. For the third selection, the probability is 1/18, and so on. This assumes that the population is **finite**—that is, the number in the population is known and relatively small in number. Examples of a finite population are 2,842 Republicans in the precinct, 9,241 applications for medical school, and the eighteen Dakota 4x4 Crew Cabs at Helfman Dodge Chrysler Jeep in Houston, Texas.

Recall that one of the criteria for the binomial distribution is that the probability of success remains the same from trial to trial. Because the probability of success does not remain the same from trial to trial when sampling is from a relatively small population without replacement, the binomial distribution should not be used. Instead, the **hypergeometric distribution** is applied. Therefore, (1) if a sample is selected from a finite population without replacement and (2) if the size of the sample n is more than 5% of the size of the population N , then the hypergeometric distribution is used to determine the probability of a specified number of successes or failures. It is especially appropriate when the size of the population is small.

The formula for the hypergeometric distribution is:

HYPERGEOMETRIC DISTRIBUTION

$$P(x) = \frac{({}_sC_x)({}_{N-s}C_{n-x})}{{}_NC_n}$$

[6-6]

where:

N is the size of the population.

S is the number of successes in the population.

x is the number of successes in the sample. It may be 0, 1, 2, 3, . . .

n is the size of the sample or the number of trials.

C is the symbol for a combination.

In summary, a hypergeometric probability distribution has these characteristics:

HYPERGEOMETRIC PROBABILITY EXPERIMENT

1. An outcome on each trial of an experiment is classified into one of two mutually exclusive categories—a success or a failure.
2. The random variable is the number of successes in a fixed number of trials.
3. The trials are not independent.
4. We assume that we sample from a finite population without replacement and $n/N > 0.05$. So, the probability of a success *changes* for each trial.

The following example illustrates the details of determining a probability using the hypergeometric distribution.

EXAMPLE

PlayTime Toys Inc. employs 50 people in the Assembly Department. Forty of the employees belong to a union and 10 do not. Five employees are selected at random to form a committee to meet with management regarding shift starting times. What is the probability that four of the five selected for the committee belong to a union?



© Howard Berman/Getty Images

SOLUTION

The population in this case is the 50 Assembly Department employees. An employee can be selected for the committee only once. Hence, the sampling is done without replacement. Thus, the probability of selecting a union employee, for example, changes from one trial to the next. The hypergeometric distribution is appropriate for determining the probability. In this problem,

N is 50, the number of employees.

S is 40, the number of union employees.

x is 4, the number of union employees selected.

n is 5, the number of employees selected.

We wish to find the probability 4 of the 5 committee members belong to a union. Inserting these values into formula (6–6):

$$P(4) = \frac{({}_{40}C_4)({}_{50-40}C_{5-4})}{{}_{50}C_5} = \frac{(91,390)(10)}{2,118,760} = .431$$

Thus, the probability of selecting 5 assembly workers at random from the 50 workers and finding 4 of the 5 are union members is .431.

Table 6–4 shows the hypergeometric probabilities of finding 0, 1, 2, 3, 4, and 5 union members on the committee.

TABLE 6–4 Hypergeometric Probabilities ($n = 5$, $N = 50$, and $S = 40$) for the Number of Union Members on the Committee

Union Members	Probability
0	.000
1	.004
2	.044
3	.210
4	.431
5	.311
	<u>1.000</u>

Table 6–5 shows a comparison of the results using the binomial distribution and the hypergeometric distribution. Because 40 of the 50 Assembly Department employees belong to the union, we let $\pi = .80$ for the binomial distribution. The binomial probabilities for Table 6–5 come from the binomial distribution with $n = 5$ and $\pi = .80$.

TABLE 6–5 Hypergeometric and Binomial Probabilities for PlayTime Toys Inc. Assembly Department

Number of Union Members on Committee	Hypergeometric Probability, $P(x)$	Binomial Probability ($n = 5$ and $\pi = .80$)
0	.000	.000
1	.004	.006
2	.044	.051
3	.210	.205
4	.431	.410
5	.311	.328
	1.000	1.000

As Table 6–5 shows, when the binomial requirement of a constant probability of success cannot be met, the hypergeometric distribution should be used. There are clear differences between the probabilities.

However, under certain conditions the results of the binomial distribution can be used to approximate the hypergeometric. This leads to a rule of thumb: if selected items are not returned to the population, the binomial distribution can be used to closely approximate the hypergeometric distribution when $n < .05N$. In other words, the binomial will closely approximate the hypergeometric distribution if the sample is less than 5% of the population. For example, if the population, N , is 150, the number of successes in the population, S , is 120, and the sample size, n , is five, then the rule of thumb is true. That is, $5 < 0.05(150)$, or $5 < 7.5$. The sample size is less than 5% of the population. In the following table, hypergeometric and binomial probability distributions are compared for this situation. The probabilities are very close.

TABLE 6–6 A Comparison of Hypergeometric and Binomial Probabilities When the Sample Size is Less than 0.05(n)

x	Hypergeometric Probability, $P(x)$	Binomial Probability ($n = 5$ and $\pi = .80 = (120/150)$)
0	.000	.000
1	.006	.006
2	.049	.051
3	.206	.205
4	.417	.410
5	.322	.328
	1.000	1.000

	A	B
1	Union Members	Probability
2	0	0.000
3	1	0.004
4	2	0.044
5	3	0.210
6	4	0.431
7	5	0.311

A hypergeometric distribution can be created using Excel. See the output for Table 6–5 on the left. The necessary steps are given in Appendix C in the back of the text.

SELF-REVIEW 6–5



Horwege Discount Brokers plans to hire five new financial analysts this year. There is a pool of 12 approved applicants, and George Horwege, the owner, decides to randomly select those who will be hired. There are eight men and four women among the approved applicants. What is the probability that three of the five hired are men?

EXERCISES

25. A CD contains 10 songs; 6 are classical and 4 are rock and roll. In a sample of three songs, what is the probability that exactly two are classical? Assume the samples are drawn without replacement.
26. A population consists of 15 items, 10 of which are acceptable. In a sample of four items, what is the probability that exactly three are acceptable? Assume the samples are drawn without replacement.
27. The Riverton Branch of the National Bank of Wyoming has 10 real estate loans over \$1,000,000. Of these 10 loans, 3 are “underwater.” A loan is underwater if the amount of the loan is greater than the value of the property. The chief loan officer decided to randomly select two of these loans to determine if they met all banking standards. What is the probability that neither of the selected loans is underwater?
28. The Computer Systems Department has eight faculty, six of whom are tenured. Dr. Vonder, the chairman, wants to establish a committee of three department faculty members to review the curriculum. If she selects the committee at random:
 - a. What is the probability all members of the committee are tenured?
 - b. What is the probability that at least one member is not tenured? (Hint: For this question, use the complement rule.)
29. Keith’s Florists has 15 delivery trucks, used mainly to deliver flowers and flower arrangements in the Greenville, South Carolina, area. Of these 15 trucks, 6 have brake problems. A sample of five trucks is randomly selected. What is the probability that two of those tested have defective brakes?
30. The game called Lotto sponsored by the Louisiana Lottery Commission pays its largest prize when a contestant matches all 6 of the 40 possible numbers. Assume there are 40 ping-pong balls each with a single number between 1 and 40. Any number appears only once, and the winning balls are selected without replacement.
 - a. The commission reports that the probability of matching all the numbers are 1 in 3,838,380. What is this in terms of probability?
 - b. Use the hypergeometric formula to find this probability.
The lottery commission also pays if a contestant matches four or five of the six winning numbers. Hint: Divide the 40 numbers into two groups, winning numbers and nonwinning numbers.
 - c. Find the probability, again using the hypergeometric formula, for matching 4 of the 6 winning numbers.
 - d. Find the probability of matching 5 of the 6 winning numbers.

LO6-6

Explain the assumptions of the Poisson distribution and apply it to calculate probabilities.

POISSON PROBABILITY DISTRIBUTION

The **Poisson probability distribution** describes the number of times some event occurs during a specified interval. Examples of an interval may be time, distance, area, or volume.

The distribution is based on two assumptions. The first assumption is that the probability is proportional to the length of the interval. The second assumption is that the intervals are independent. To put it another way, the longer the interval, the larger the probability, and the number of occurrences in one interval does not affect the other intervals. This distribution is a limiting form of the binomial distribution when the probability of a success is very small and n is large. It is often referred to as the “law of improbable events,” meaning that the probability, π , of a particular event’s happening is quite small. The Poisson distribution is a discrete probability distribution because it is formed by counting.

STATISTICS IN ACTION

Near the end of World War II, the Germans developed rocket bombs, which were fired at the city of London. The Allied military command didn't know whether these bombs were fired at random or whether they had an aiming device. To investigate, the city of London was divided into 586 square regions. The distribution of hits in each square was recorded as follows:

Hits	0	1	2	3	4	5
Regions	229	221	93	35	7	1

To interpret, the above chart indicates that 229 regions were not hit with one of the bombs. Seven regions were hit four times. Using the Poisson distribution, with a mean of 0.93 hits per region, the expected number of hits is as follows:

Hits	0	1	2	3	4	5 or more
Regions	231.2	215.0	100.0	31.0	7.2	1.6

Because the actual number of hits was close to the expected number of hits, the military command concluded that the bombs were falling at random. The Germans had not developed a bomb with an aiming device.

The Poisson probability distribution has these characteristics:

POISSON PROBABILITY EXPERIMENT

- 1. The random variable is the number of times some event occurs during a defined interval.
- 2. The probability of the event is proportional to the size of the interval.
- 3. The intervals do not overlap and are independent.

This probability distribution has many applications. It is used as a model to describe the distribution of errors in data entry, the number of scratches and other imperfections in newly painted car panels, the number of defective parts in outgoing shipments, the number of customers waiting to be served at a restaurant or waiting to get into an attraction at Disney World, and the number of accidents on I-75 during a three-month period.

The Poisson distribution is described mathematically by the formula:

POISSON DISTRIBUTION

$$P(x) = \frac{\mu^x e^{-\mu}}{x!} \tag{6-7}$$

where:

- μ (mu) is the mean number of occurrences (successes) in a particular interval.
- e is the constant 2.71828 (base of the Napierian logarithmic system).
- x is the number of occurrences (successes).
- $P(x)$ is the probability for a specified value of x .

The mean number of successes, μ , is found by $n\pi$, where n is the total number of trials and π the probability of success.

MEAN OF A POISSON DISTRIBUTION

$$\mu = n\pi \tag{6-8}$$

The variance of the Poisson is equal to its mean. If, for example, the probability that a check cashed by a bank will bounce is .0003, and 10,000 checks are cashed, the mean and the variance for the number of bad checks is 3.0, found by $\mu = n\pi = 10,000(.0003) = 3.0$.

Recall that for a binomial distribution there are a fixed number of trials. For example, for a four-question multiple-choice test there can only be zero, one, two, three, or four successes (correct answers). The random variable, x , for a Poisson distribution, however, can assume an *infinite number of values*—that is, 0, 1, 2, 3, 4, 5, . . . However, *the probabilities become very small after the first few occurrences* (successes).

EXAMPLE

Budget Airlines is a seasonal airline that operates flights from Myrtle Beach, South Carolina, to various cities in the northeast. The destinations include Boston, Pittsburgh, Buffalo, and both LaGuardia and JFK airports in New York City. Recently Budget has been concerned about the number of lost bags. Ann Poston from the Analytics Department was asked to study the issue. She randomly selected a sample of 500 flights and found that a total of twenty bags were lost on the sampled flights.

Show that this situation follows the Poisson distribution. What is the mean number of bags lost per flight? What is the likelihood that no bags are lost on a flight? What is the probability at least one bag is lost?

SOLUTION

To begin, let's confirm that the Budget Airlines situation follows a Poisson Distribution. Refer to the highlighted box labeled Poisson Probability Experiment in this section. We count the number of bags lost on a particular flight. On most flights there were no bags lost, on a few flights one was lost, and perhaps in very rare circumstances more than one bag was lost. The continuum or interval is a particular flight. Each flight is assumed to be independent of any other flight.

Based on the sample information we can estimate the mean number of bags lost per flight. There were 20 bags lost in 500 flights so the mean number of bags lost per flight is .04, found by $20/500$. Hence $\mu = .04$.

We use formula (6–7) to find the probability of any number of lost bags. In this case x , the number of lost bags is 0.

$$P(0) = \frac{\mu^x e^{-\mu}}{x!} = \frac{.04^0 e^{-.04}}{0!} = .9608$$

The probability of exactly one lost bag is:

$$P(1) = \frac{\mu^x e^{-\mu}}{x!} = \frac{.04^1 e^{-.04}}{1!} = .0384$$

The probability of one or more lost bags is:

$$1 - P(0) = 1 - \frac{\mu^x e^{-\mu}}{x!} = 1 - \frac{.04^0 e^{-.04}}{0!} = 1 - .9608 = .0392$$

These probabilities can also be found using Excel. The commands to compute Poisson probabilities are in Appendix C.

	A	B
1	Success	Probability
2	0	0.9608
3	1	0.0384
4	2	0.0008
5	3	0.0000
6	4	0.0000
7	5	0.0000
8	6	0.0000
9	7	0.0000

Part of Appendix B.2 is repeated as Table 6–7. For certain values of μ , the mean of the Poisson distribution, we can read the probability directly from the table. Turning to another example, NewYork-LA Trucking Company finds the mean number of breakdowns on the New York to Los Angeles route is 0.30. From Table 6–7 we can locate the probability of no breakdowns on a particular run. First find the column headed “0.30” then read down that column to the row labeled “0”. The value at the intersection is .7408, so this value is the probability of no breakdowns on a particular run. The probability of one breakdown is .2222.

TABLE 6–7 Poisson Table for Various Values of μ (from Appendix B.2)

x	μ								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	0.9048	0.8187	0.7408	0.6703	0.6065	0.5488	0.4966	0.4493	0.4066
1	0.0905	0.1637	0.2222	0.2681	0.3033	0.3293	0.3476	0.3595	0.3659
2	0.0045	0.0164	0.0333	0.0536	0.0758	0.0988	0.1217	0.1438	0.1647
3	0.0002	0.0011	0.0033	0.0072	0.0126	0.0198	0.0284	0.0383	0.0494
4	0.0000	0.0001	0.0003	0.0007	0.0016	0.0030	0.0050	0.0077	0.0111
5	0.0000	0.0000	0.0000	0.0001	0.0002	0.0004	0.0007	0.0012	0.0020
6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0002	0.0003
7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Earlier in this section, we mentioned that the Poisson probability distribution is a limiting form of the binomial. That is, we could estimate a binomial probability using the Poisson. In the following example, we use the Poisson distribution to estimate a binomial probability when n , the number of trials, is large and π , the probability of a success, small.

▶ **EXAMPLE**

Coastal Insurance Company underwrites insurance for beachfront properties along the Virginia, North and South Carolina, and Georgia coasts. It uses the estimate that the probability of a named Category III hurricane (sustained winds of more than 110 miles per hour) or higher striking a particular region of the coast (for example, St. Simons Island, Georgia) in any one year is .05. If a homeowner takes a 30-year mortgage on a recently purchased property in St. Simons, what is the likelihood that the owner will experience at least one hurricane during the mortgage period?

SOLUTION

To use the Poisson probability distribution, we begin by determining the mean or expected number of storms meeting the criterion hitting St. Simons during the 30-year period. That is:

$$\mu = n\pi = 30(.05) = 1.5$$

where:

- n is the number of years, 30 in this case.
- π is the probability a hurricane meeting the strength criteria comes ashore.
- μ is the mean or expected number of storms in a 30-year period.

To find the probability of at least one storm hitting St. Simons Island, Georgia, we first find the probability of no storms hitting the coast and subtract that value from 1.

$$P(x \geq 1) = 1 - P(x = 0) = 1 - \frac{\mu^0 e^{-1.5}}{0!} = 1 - .2231 = .7769$$

We conclude that the likelihood a hurricane meeting the strength criteria will strike the beachfront property at St. Simons during the 30-year period when the mortgage is in effect is .7769. To put it another way, the probability St. Simons will be hit by a Category III or higher hurricane during the 30-year period is a little more than 75%.

We should emphasize that the continuum, as previously described, still exists. That is, there are expected to be 1.5 storms hitting the coast per 30-year period. The continuum is the 30-year period.

In the preceding case, we are actually using the Poisson distribution as an estimate of the binomial. Note that we've met the binomial conditions outlined on page 183.

- There are only two possible outcomes: a hurricane hits the St. Simons area or it does not.
- There are a fixed number of trials, in this case 30 years.
- There is a constant probability of success; that is, the probability of a hurricane hitting the area is .05 each year.
- The years are independent. That means if a named storm strikes in the fifth year, that has no effect on any other year.

To find the probability of at least one storm striking the area in a 30-year period using the binomial distribution:

$$P(x \geq 1) = 1 - P(x = 0) = 1 - [{}_{30}C_0(.05)^0(.95)^{30}] = 1 - [(1)(1)(.2146)] = .7854$$

The probability of at least one hurricane hitting the St. Simons area during the 30-year period using the binomial distribution is .7854.

Which answer is correct? Why should we look at the problem both ways? The binomial is the more “technically correct” solution. The Poisson can be thought of as an approximation for the binomial, when n , the number of trials is large, and π , the probability of a success, is small. We look at the problem using both distributions to emphasize the convergence of the two discrete distributions. In some instances, using the Poisson may be the quicker solution, and as you see there is little practical difference in the answers. In fact, as n gets larger and π smaller, the difference between the two distributions gets smaller.

The Poisson probability distribution is always positively skewed and the random variable has no specific upper limit. In the lost bags example/solution, the Poisson distribution, with $\mu = 0.04$, is highly skewed. As μ becomes larger, the Poisson distribution becomes more symmetrical. For example, Chart 6–4 shows the distributions of

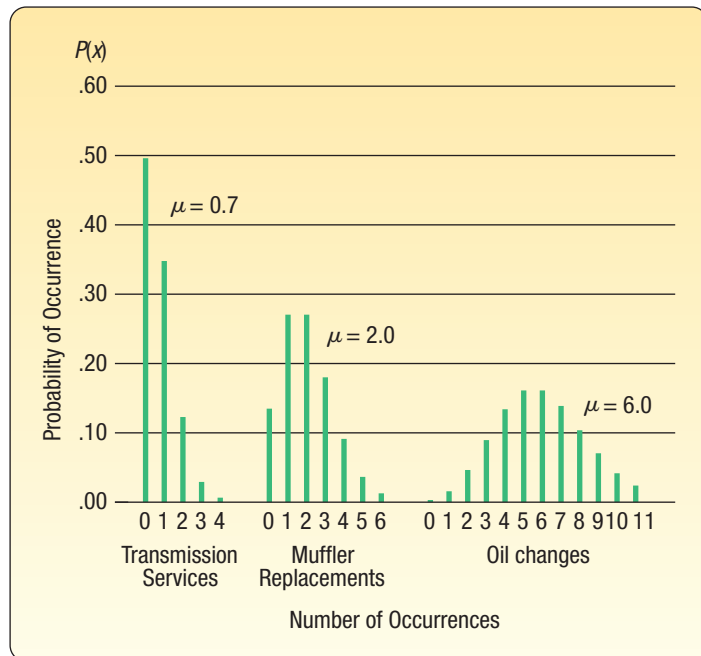


CHART 6–4 Poisson Probability Distributions for Means of 0.7, 2.0, and 6.0

the number of transmission services, muffler replacements, and oil changes per day at Avellino's Auto Shop. They follow Poisson distributions with means of 0.7, 2.0, and 6.0, respectively.

In summary, the Poisson distribution is a family of discrete distributions. All that is needed to construct a Poisson probability distribution is the mean number of defects, errors, or other random variable, designated as μ .

SELF-REVIEW 6-6



From actuary tables, Washington Insurance Company determined the likelihood that a man age 25 will die within the next year is .0002. If Washington Insurance sells 4,000 policies to 25-year-old men this year, what is the probability they will pay on exactly one policy?

EXERCISES

31. In a Poisson distribution $\mu = 0.4$.
 - a. What is the probability that $x = 0$?
 - b. What is the probability that $x > 0$?
32. In a Poisson distribution $\mu = 4$.
 - a. What is the probability that $x = 2$?
 - b. What is the probability that $x \leq 2$?
 - c. What is the probability that $x > 2$?
33. Ms. Bergen is a loan officer at Coast Bank and Trust. From her years of experience, she estimates that the probability is .025 that an applicant will not be able to repay his or her installment loan. Last month she made 40 loans.
 - a. What is the probability that three loans will be defaulted?
 - b. What is the probability that at least three loans will be defaulted?
34. Automobiles arrive at the Elkhart exit of the Indiana Toll Road at the rate of two per minute. The distribution of arrivals approximates a Poisson distribution.
 - a. What is the probability that no automobiles arrive in a particular minute?
 - b. What is the probability that at least one automobile arrives during a particular minute?
35. It is estimated that 0.5% of the callers to the Customer Service department of Dell Inc. will receive a busy signal. What is the probability that of today's 1,200 callers at least 5 received a busy signal?
36. In the past, schools in Los Angeles County have closed an average of 3 days each year for weather emergencies. What is the probability that schools in Los Angeles County will close for 4 days next year?

CHAPTER SUMMARY

- I. A random variable is a numerical value determined by the outcome of an experiment.
- II. A probability distribution is a listing of all possible outcomes of an experiment and the probability associated with each outcome.
 - A. A discrete probability distribution can assume only certain values. The main features are:
 1. The sum of the probabilities is 1.00.
 2. The probability of a particular outcome is between 0.00 and 1.00.
 3. The outcomes are mutually exclusive.
 - B. A continuous distribution can assume an infinite number of values within a specific range.
- III. The mean and variance of a probability distribution are computed as follows.
 - A. The mean is equal to:

$$\mu = \sum[xP(x)] \quad (6-1)$$

- B. The variance is equal to:

$$\sigma^2 = \sum[(x - \mu)^2P(x)] \quad (6-2)$$

IV. The binomial distribution has the following characteristics.

- A.** Each outcome is classified into one of two mutually exclusive categories.
- B.** The distribution results from a count of the number of successes in a fixed number of trials.
- C.** The probability of a success remains the same from trial to trial.
- D.** Each trial is independent.
- E.** A binomial probability is determined as follows:

$$P(x) = {}_nC_x \pi^x (1 - \pi)^{n-x} \quad (6-3)$$

- F.** The mean is computed as:

$$\mu = n\pi \quad (6-4)$$

- G.** The variance is

$$\sigma^2 = n\pi(1 - \pi) \quad (6-5)$$

V. The hypergeometric distribution has the following characteristics.

- A.** There are only two possible outcomes.
- B.** The probability of a success is not the same on each trial.
- C.** The distribution results from a count of the number of successes in a fixed number of trials.
- D.** It is used when sampling without replacement from a finite population.
- E.** A hypergeometric probability is computed from the following equation:

$$P(x) = \frac{{}_sC_x {}_{N-s}C_{n-x}}{{}_NC_n} \quad (6-6)$$

VI. The Poisson distribution has the following characteristics.

- A.** It describes the number of times some event occurs during a specified interval.
- B.** The probability of a “success” is proportional to the length of the interval.
- C.** Nonoverlapping intervals are independent.
- D.** It is a limiting form of the binomial distribution when n is large and π is small.
- E.** A Poisson probability is determined from the following equation:

$$P(x) = \frac{\mu^x e^{-\mu}}{x!} \quad (6-7)$$

- F.** The mean and the variance are:

$$\begin{aligned} \mu &= n\pi \\ \sigma^2 &= n\pi \end{aligned} \quad (6-8)$$

CHAPTER EXERCISES

- 37.** What is the difference between a random variable and a probability distribution?
- 38.** For each of the following indicate whether the random variable is discrete or continuous.
 - a.** The length of time to get a haircut.
 - b.** The number of cars a jogger passes each morning while running.
 - c.** The number of hits for a team in a high school girls' softball game.
 - d.** The number of patients treated at the South Strand Medical Center between 6 and 10 p.m. each night.
 - e.** The distance your car traveled on the last fill-up.
 - f.** The number of customers at the Oak Street Wendy's who used the drive-through facility.
 - g.** The distance between Gainesville, Florida, and all Florida cities with a population of at least 50,000.
- 39.** **FILE** An investment will be worth \$1,000, \$2,000, or \$5,000 at the end of the year. The probabilities of these values are .25, .60, and .15, respectively. Determine the mean and variance of the investment's dollar value.

40. The following notice appeared in the golf shop at a Myrtle Beach, South Carolina, golf course.

Blackmoor Golf Club Members

The golf shop is holding a raffle to win a TaylorMade M1 10.5° Regular Flex Driver (\$300 value).
Tickets are \$5.00 each.
Only 80 tickets will be sold.
Please see the golf shop to get your tickets!

- John Underpar buys a ticket.
- a. What are Mr. Underpar’s possible monetary outcomes?
 - b. What are the probabilities of the possible outcomes?
 - c. Summarize Mr. Underpar’s “experiment” as a probability distribution.
 - d. What is the mean or expected value of the probability distribution? Explain your result.
 - e. If all 80 tickets are sold, what is the expected return to the Club?
41. **FILE** Croissant Bakery Inc. offers special decorated cakes for birthdays, weddings, and other occasions. It also has regular cakes available in its bakery. The following table gives the total number of cakes sold per day and the corresponding probability. Compute the mean, variance, and standard deviation of the number of cakes sold per day.

Number of Cakes Sold in a Day	Probability
12	.25
13	.40
14	.25
15	.10

42. **FILE** The payouts for the Powerball lottery and their corresponding odds and probabilities of occurrence are shown below. The price of a ticket is \$1.00. Find the mean and standard deviation of the payout. Hint: Don’t forget to include the cost of the ticket and its corresponding probability.

Divisions	Payout	Odds	Probability
Five plus Powerball	\$50,000,000	146,107,962	0.000000006844
Match 5	200,000	3,563,609	0.000000280614
Four plus Powerball	10,000	584,432	0.000001711060
Match 4	100	14,255	0.000070145903
Three plus Powerball	100	11,927	0.000083836351
Match 3	7	291	0.003424657534
Two plus Powerball	7	745	0.001340482574
One plus Powerball	4	127	0.007812500000
Zero plus Powerball	3	69	0.014285714286

43. In a recent study, 35% of people surveyed indicated chocolate was their favorite flavor of ice cream. Suppose we select a sample of 10 people and ask them to name their favorite flavor of ice cream.
- a. How many of those in the sample would you expect to name chocolate?
 - b. What is the probability exactly four of those in the sample name chocolate?
 - c. What is the probability four or more name chocolate?
44. **FILE** Thirty percent of the population in a southwestern community are Spanish-speaking Americans. A Spanish-speaking person is accused of killing a non-Spanish-speaking American and goes to trial. Of the first 12 potential jurors, only 2 are Spanish-speaking Americans, and 10 are not. The defendant’s lawyer challenges the jury selection, claiming bias against her client. The government lawyer disagrees, saying that the probability of this particular jury composition is common. Compute the probability and discuss the assumptions.

45. An auditor for Health Maintenance Services of Georgia reports 40% of policyholders 55 years or older submit a claim during the year. Fifteen policyholders are randomly selected for company records.
- How many of the policyholders would you expect to have filed a claim within the last year?
 - What is the probability that 10 of the selected policyholders submitted a claim last year?
 - What is the probability that 10 or more of the selected policyholders submitted a claim last year?
 - What is the probability that more than 10 of the selected policyholders submitted a claim last year?
46. Tire and Auto Supply is considering a 2-for-1 stock split. Before the transaction is finalized, at least two-thirds of the 1,200 company stockholders must approve the proposal. To evaluate the likelihood the proposal will be approved, the CFO selected a sample of 18 stockholders. He contacted each and found 14 approved of the proposed split. What is the likelihood of this event, assuming two-thirds of the stockholders approve?
47. A federal study reported that 7.5% of the U.S. workforce has a drug problem. A drug enforcement official for the state of Indiana wished to investigate this statement. In her sample of 20 employed workers:
- How many would you expect to have a drug problem? What is the standard deviation?
 - What is the likelihood that *none* of the workers sampled has a drug problem?
 - What is the likelihood *at least one* has a drug problem?
48. The Bank of Hawaii reports that 7% of its credit card holders will default at some time in their life. The Hilo branch just mailed out 12 new cards today.
- How many of these new cardholders would you expect to default? What is the standard deviation?
 - What is the likelihood that *none* of the cardholders will default?
 - What is the likelihood *at least one* will default?
49. Recent statistics suggest that 15% of those who visit a retail site on the internet to make a purchase. A retailer wished to verify this claim. To do so, she selected a sample of 16 “hits” to her site and found that 4 had actually made a purchase.
- What is the likelihood of exactly four purchases?
 - How many purchases should she expect?
 - What is the likelihood that four or more “hits” result in a purchase?
50. In Chapter 19, we discuss *acceptance sampling*. Acceptance sampling is a statistical method used to monitor the quality of purchased parts and components. To ensure the quality of incoming parts, a purchaser or manufacturer normally samples 20 parts and allows one defect.
- What is the likelihood of accepting a lot that is 1% defective?
 - If the quality of the incoming lot was actually 2%, what is the likelihood of accepting it?
 - If the quality of the incoming lot was actually 5%, what is the likelihood of accepting it?
51. Unilever Inc. recently developed a new body wash with a scent of ginger. Their research indicates that 30% of men like the new scent. To further investigate, Unilever’s marketing research group randomly selected 15 men and asked them if they liked the scent. What is the probability that six or more men like the ginger scent in the body wash?
52. Dr. Richmond, a psychologist, is studying the daytime television viewing habits of college students. She believes 45% of college students watch soap operas during the afternoon. To further investigate, she selects a sample of 10.
- Develop a probability distribution for the number of students in the sample who watch soap operas.
 - Find the mean and the standard deviation of this distribution.
 - What is the probability of finding exactly four students who watch soap operas?
 - What is the probability less than half of the students selected watch soap operas?
53. **FILE** A recent study conducted by Penn, Shone, and Borland, on behalf of LastMinute.com, revealed that 52% of business travelers plan their trips less than two weeks before departure. The study is to be replicated in the tri-state area with a sample of 12 frequent business travelers.
- Develop a probability distribution for the number of travelers who plan their trips within two weeks of departure.
 - Find the mean and the standard deviation of this distribution.

- c. What is the probability exactly 5 of the 12 selected business travelers plan their trips within two weeks of departure?
 - d. What is the probability 5 or fewer of the 12 selected business travelers plan their trips within two weeks of departure?
54. The Internal Revenue Service is studying the category of charitable contributions. A sample of 25 returns is selected from young couples between the ages of 20 and 35 who had an adjusted gross income of more than \$100,000. Of these 25 returns, five had charitable contributions of more than \$1,000. Four of these returns are selected for a comprehensive audit.
- a. Explain why the hypergeometric distribution is appropriate.
 - b. What is the probability exactly one of the four audited had a charitable deduction of more than \$1,000?
 - c. What is the probability at least one of the audited returns had a charitable contribution of more than \$1,000?
55. The law firm of Hagel and Hagel is located in downtown Cincinnati. There are 10 partners in the firm; 7 live in Ohio and 3 in northern Kentucky. Ms. Wendy Hagel, the managing partner, wants to appoint a committee of 3 partners to look into moving the firm to northern Kentucky. If the committee is selected at random from the 10 partners, what is the probability that:
- a. One member of the committee lives in northern Kentucky and the others live in Ohio?
 - b. At least one member of the committee lives in northern Kentucky?
56. Topen is a leading source on energy-efficient products. Their list of the top seven vehicles in terms of fuel efficiency for 2017 includes three Hondas.
- a. Determine the probability distribution for the number of Hondas in a sample of two cars chosen from the top seven.
 - b. What is the likelihood that in the sample of two at least one Honda is included?
57. The position of chief of police in the city of Corry, Pennsylvania, is vacant. A search committee of Corry residents is charged with the responsibility of recommending a new chief to the city council. There are 12 applicants, 4 of whom are either female or members of a minority. The search committee decides to interview all 12 of the applicants. To begin, they randomly select four applicants to be interviewed on the first day, and none of the four is female or a member of a minority. The local newspaper, the *Corry Press*, suggests discrimination in an editorial. What is the likelihood of this occurrence?
58. **FILE** Listed below is the population by state for the 15 states with the largest population. Also included is whether that state's border touches the Gulf of Mexico, the Atlantic Ocean, or the Pacific Ocean (coastline).

Rank	State	Population	Coastline
1	California	38,802,500	Yes
2	Texas	26,956,958	Yes
3	Florida	19,893,297	Yes
4	New York	19,746,227	Yes
5	Illinois	12,880,580	No
6	Pennsylvania	12,787,209	No
7	Ohio	11,594,163	No
8	Georgia	10,097,343	Yes
9	North Carolina	9,943,964	Yes
10	Michigan	9,909,877	No
11	New Jersey	8,938,175	Yes
12	Virginia	8,326,289	Yes
13	Washington	7,061,530	Yes
14	Massachusetts	6,745,408	Yes
15	Arizona	6,731,484	No

Note that 5 of the 15 states do not have any coastline. Suppose three states are selected at random. What is the probability that:

- a. None of the states selected has any coastline?
- b. Exactly one of the selected states has a coastline?
- c. At least one of the selected states has a coastline?

59. The sales of Lexus automobiles in the Detroit area follow a Poisson distribution with a mean of 3 per day.
- What is the probability that no Lexus is sold on a particular day?
 - What is the probability that for 5 consecutive days at least one Lexus is sold?
60. Suppose 1.5% of the antennas on new Nokia cell phones are defective. For a random sample of 200 antennas, find the probability that:
- None of the antennas is defective.
 - Three or more of the antennas are defective.
61. A study of the checkout lines at the Safeway Supermarket in the South Strand area revealed that between 4 and 7 p.m. on weekdays there is an average of four customers waiting in line. What is the probability that you visit Safeway today during this period and find:
- No customers are waiting?
 - Four customers are waiting?
 - Four or fewer are waiting?
 - Four or more are waiting?
62. An internal study by the Technology Services department at Lahey Electronics revealed company employees receive an average of two non-work-related e-mails per hour. Assume the arrival of these e-mails is approximated by the Poisson distribution.
- What is the probability Linda Lahey, company president, received exactly one non-work-related e-mail between 4 p.m. and 5 p.m. yesterday?
 - What is the probability she received five or more non-work-related e-mails during the same period?
 - What is the probability she did not receive any non-work-related e-mails during the period?
63. Recent crime reports indicate that 3.1 motor vehicle thefts occur each minute in the United States. Assume that the distribution of thefts per minute can be approximated by the Poisson probability distribution.
- Calculate the probability exactly *four* thefts occur in a minute.
 - What is the probability there are *no* thefts in a minute?
 - What is the probability there is *at least one* theft in a minute?
64. Recent difficult economic times have caused an increase in the foreclosure rate of home mortgages. Statistics from the Penn Bank and Trust Company show their monthly foreclosure rate is now 1 loan out of every 136 loans. Last month the bank approved 300 loans.
- How many foreclosures would you expect the bank to have last month?
 - What is the probability of exactly two foreclosures?
 - What is the probability of at least one foreclosure?
65. The National Aeronautics and Space Administration (NASA) has experienced two disasters. The *Challenger* exploded over the Atlantic Ocean in 1986, and the *Columbia* disintegrated on reentry over East Texas in 2003. Based on the first 113 missions, and assuming failures occur at the same rate, consider the next 23 missions. What is the probability of exactly two failures? What is the probability of no failures?
66. According to the “January theory,” if the stock market is up for the month of January, it will be up for the year. If it is down in January, it will be down for the year. According to an article in *The Wall Street Journal*, this theory held for 29 out of the last 34 years. Suppose there is no truth to this theory; that is, the probability it is either up or down is .50. What is the probability this could occur by chance? You will probably need a software package such as Excel or Minitab.
67. During the second round of the 1989 U.S. Open golf tournament, four golfers scored a hole in one on the sixth hole. The odds of a professional golfer making a hole in one are estimated to be 3,708 to 1, so the probability is $1/3,709$. There were 155 golfers participating in the second round that day. Estimate the probability that four golfers would score a hole in one on the sixth hole.
68. According to sales information in the first quarter of 2016, 2.7% of new vehicles sold in the United States were hybrids. This is down from 3.3% for the same period a year earlier. An analyst’s review of the data indicates that the reasons for the sales decline include the low price of gasoline and the higher price of a hybrid compared to similar vehicles. Let’s assume these statistics remain the same for 2017. That is, 2.7 percent of

new car sales are hybrids in the first quarter of 2017. For a sample of 40 vehicles sold in the Richmond, Virginia area:

- a. How many vehicles would you expect to be hybrid?
 - b. Use the Poisson distribution to find the probability that five of the sales were hybrid vehicles.
 - c. Use the binomial distribution to find the probability that five of the sales were hybrid vehicles.
69. A recent CBS News survey reported that 67% of adults felt the U.S. Treasury should continue making pennies. Suppose we select a sample of 15 adults.
- a. How many of the 15 would we expect to indicate that the Treasury should continue making pennies? What is the standard deviation?
 - b. What is the likelihood that exactly eight adults would indicate the Treasury should continue making pennies?
 - c. What is the likelihood at least eight adults would indicate the Treasury should continue making pennies?

DATA ANALYTICS

70. **FILE** Refer to the North Valley Real Estate data, which report information on homes sold in the area last year.
- a. Create a probability distribution for the number of bedrooms. Compute the mean and the standard deviation of this distribution.
 - b. Create a probability distribution for the number of bathrooms. Compute the mean and the standard deviation of this distribution.
71. **FILE** Refer to the Baseball 2016 data. Compute the mean number of home runs per game. To do this, first find the mean number of home runs per team for 2016. Next, divide this value by 162 (a season comprises 162 games). Then multiply by 2 because there are two teams in each game. Use the Poisson distribution to estimate the number of home runs that will be hit in a game. Find the probability that:
- a. There are no home runs in a game.
 - b. There are two home runs in a game.
 - c. There are at least four home runs in a game.